

The Lennard-Jones Fluid

Potential, Reduced Units & the State Point

Monte Carlo Simulations — Master's Course

The Lennard-Jones Potential

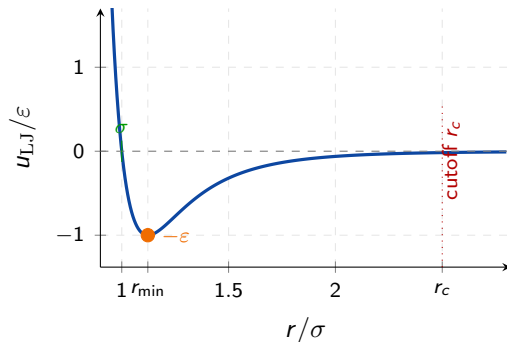
Formula

$$u_{\text{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

- r — interparticle distance
- ε — well depth (energy scale)
- σ — collision diameter (length scale)

r^{-12} **Pauli repulsion** (electron-cloud overlap)

r^{-6} **London dispersion** (induced-dipole attraction)



Cut-and-shifted potential

$$r_c = 2.5\sigma: \quad u(r) = \begin{cases} u_{\text{LJ}}(r) - u_{\text{LJ}}(r_c) & r \leq r_c \\ 0 & r > r_c \end{cases}$$

Reduced (Dimensionless) Units

Definition

Set $\varepsilon = \sigma = k_B = 1$. Every quantity is a *multiple of the natural LJ scale*.

Quantity	Reduced	SI unit
Length	$r^* = r/\sigma$	m
Energy	$u^* = u/\varepsilon$	J
Temperature	$T^* = k_B T/\varepsilon$	K
Density	$\rho^* = \rho \sigma^3$	m^{-3}
Pressure	$P^* = P \sigma^3/\varepsilon$	Pa
Time	$t^* = t \sqrt{\varepsilon/m\sigma^2}$	s

Why? Code has no units — results rescale to any LJ substance by choosing ε and σ .

Example: Liquid Argon

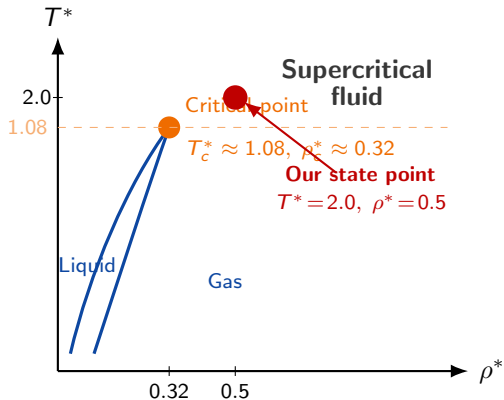
$$\sigma = 3.405 \text{ \AA}, \quad \varepsilon/k_B = 119.8 \text{ K}, \quad m = 39.95 \text{ u}$$

Reduced value	Physical value
$T^* = 2.0$	$T = 2.0 \times 119.8 \approx 240 \text{ K}$
$\rho^* = 0.5$	$\rho \approx 1.26 \text{ g cm}^{-3}$
$r_c^* = 2.5$	$r_c = 8.51 \text{ \AA}$
$P^* = 1.34$	$P \approx 390 \text{ MPa}$

Our simulation state point

$T^* = 2.0, \quad \rho^* = 0.5$
Supercritical fluid ($T_c^* \approx 1.08, \rho_c^* \approx 0.32$)
Single fluid phase — no liquid/gas coexistence

Where Is Our State Point?



Key facts about our state point

- $T^* = 2.0 \approx 1.85 T_c^*$ — well above critical
- Single fluid phase throughout the simulation
- Dense fluid ($\rho^* = 0.5$): atoms frequently interact
- Reference EOS (Thol *et al.*, 2015) for comparison

EOS reference values at $T^* = 2.0$, $\rho^* = 0.5$

Pressure P^*	1.338
Potential energy u^*	-2.884
c_V/k_B	2.32